CAM-IMX585-Mono/CAM-IMX585-Color MIPI Camera Module

1. Product Overview

The **CAM-IMX585-Mono** and **CAM-IMX585-Color** are high-performance CMOS image sensors designed for demanding imaging applications in embedded systems. The

Mono variant features a monochrome sensor for enhanced low-light sensitivity, while the **Color** variant provides full RGB color imaging. Both feature an advanced Starvis 2 back-illuminated pixel structure, delivering 8.3MP resolution with exceptional low-light performance, high dynamic range, and precise image quality across diverse lighting conditions.

With native 3840×2160 (4K UHD) resolution and support for 10/12/16-bit RAW output, the CAM-IMX585-Mono/Color enables professional-grade imaging for surveillance, industrial inspection, machine vision, and embedded vision applications. The sensor's MIPI CSI-2 4-lane interface ensures reliable high-speed data transmission to host processors like Raspberry Pi 5 and NVIDIA Jetson. The on-board **FT24C08A EEPROM** (1 KB) supports camera calibration data storage and I2C-based read/write operations.

Note: While the sensor supports 10/12/16-bit RAW output, the current driver configuration operates in 12-bit RAW (R12_CSI2P) mode on Raspberry Pi 5.

1.1 Key Features

- **8.3MP 4K resolution** with Starvis 2 technology
- **Monochrome (Mono)** sensor for enhanced low-light sensitivity / **Color (RGB)** sensor for full-spectrum imaging
- **MIPI CSI-2 4-lane** high-speed interface
- **10/12/16-bit RAW output capability** (currently 12-bit)
- **Excellent low-light performance** with back-illuminated pixels
- **On-board EEPROM** (FT24C08A) for calibration data storage
- **I2C interface** for EEPROM and sensor register access
- **Compatible with Raspberry Pi**

1.2 Industry Applications

- Security & Surveillance (24/7 monitoring)
- Industrial & Machine Vision (AOI, robotics)
- Broadcast & ProAV (live streaming, drones)
- Scientific & Medical (microscopy, diagnostics)
- Automotive (ADAS, surround-view cameras)

2. Hardware Specifications

2.1 Sensor Specifications

Parameter	Specification
Product Model	CAM-IMX585-Mono / CAM-IMX585-Color
Sensor Model	Sony IMX585 (Monochrome) / Sony IMX585 (Color)
Sensor Type	CMOS (Mono) / CMOS (Color)
Pixel Technology	Starvis 2
Effective Resolution	8.3MP
Optical Format	$^{1}/_{1}$.2”
Pixel Size	2.9µm × 2.9µm
Active Pixels	3840 (H) × 2160 (V)
Max Resolution	3840×2160 (4K UHD)
Supported RAW Output	10/12/16-bit

2.2 Video Format & Resolution/Frame Rate

Resolution	Frame Rate	Pixel Format	Notes
1928×1090	30.00 fps	R12_CSI2P	Cropped from 4K
3856×2180	30.00 fps	R12_CSI2P	Native sensor readout

2.3 Interface & Physical Specifications

Parameter	Specification
Interface Type	MIPI CSI-2
Data Lanes	4-lane
Data Rate	1.5 Gbps per lane
Connector	22-pin FPC
Dimensions	38mm × 38mm
Lens Mount	CS-mount / M12

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3. Software Installation

This repository provides the necessary drivers, Image Processing Algorithm (IPA) modules, and an automated installation script to enable full support for the IMX585 sensor on Raspberry Pi.

3.1 Repository Contents

- `pkg1-imx585-driver-6.12y-offline.tar.gz` : Offline driver build package.
- `pkg2-rpicam-libcamera-offline.tar.gz` : Offline libcamera build package.

3.2 Step 1: Offline Driver Compilation

For advanced users who need to compile the driver from source:

Package: `pkg1-imx585-driver-6.12y-offline.tar.gz`

Contents:

- `imx585-v4l2-driver/` - Kernel driver source code
- `install.sh` - Automated driver installation script

Installation:

```
$ tar -xzf pkg1-imx585-driver-6.12y-offline.tar.gz
$ cd pkg1-imx585-driver
$ chmod +x install.sh
$ sudo ./install.sh
```

3.3 Step 2: Offline libcamera & rpicam-apps Compilation

For complete offline compilation of libcamera with IMX585 support and rpicam-apps:

Package: `pkg2-rpicam-libcamera-offline.tar.gz`

Contents:

- `libcamera-imx585/` - libcamera source with IMX585 IPA support
- `rpicam-apps-imx585/` - rpicam-apps source
- `build.sh` - Automated build and installation script

Installation:

```
$ tar -xzf pkg2-rpicam-libcamera-offline.tar.gz
$ cd pkg2-rpicam-offline
$ chmod +x build.sh
$ sudo ./build.sh           # Full mode with Qt support
$ sudo ./build.sh --lite    # Lite mode (minimal dependencies)
```

Build Time: ~30-40 minutes (full mode) or ~15-20 minutes (lite mode)

3.4 Manual Configuration

Edit your `/boot/firmware/config.txt` (Pi 5) or `/boot/config.txt` (Pi 4) and add one of the following configurations:

Default (CAM1 port, Color mode) - for CAM-IMX585-Color:

```
camera_auto_detect=0
dtoverlay=imx585
```

Default (CAM1 port, Monochrome mode) - for CAM-IMX585-Mono:

```
camera_auto_detect=0
dtoverlay=imx585,mono
```

Use CAM0 port (Color):

```
dtoverlay=imx585,cam0
```

Use CAM0 port (Monochrome):

```
dtoverlay=imx585,cam0,mono
```

Reboot your Raspberry Pi for the changes to take effect:

```
$ sudo reboot
```

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4. I2C Tools & EEPROM Access

The camera module includes an on-board **FT24C08A EEPROM** (1 KB total) for storing calibration data and camera parameters. The `i2c-tools/` directory provides a Python-based utility for reading, writing, and managing EEPROM data over I2C.

4.1 EEPROM Overview

The EEPROM consists of four 256-byte pages at I2C addresses **0x50, 0x51, 0x52, 0x53**, providing 1 KB of persistent storage. This is useful for:

- Storing camera calibration coefficients
- Saving sensor configuration parameters
- Preserving user-defined settings across power cycles

4.2 Using i2c-tools

The `i2c-tools/` folder contains:

- `i2c.py` - Python 3 utility (stdlib only, no external dependencies)
- `README.md` - Detailed usage documentation

Quick Start:

```
# Make the script executable
chmod +x i2c-tools/i2c.py

# Detect EEPROM chips on I2C bus 4 (CAM1 port on Pi5)
sudo python3 i2c-tools/i2c.py eeprom detect --bus 4

# Backup EEPROM to a file (recommended before any writes)
sudo python3 i2c-tools/i2c.py eeprom dump --bus 4 --out eeprom_backup.bin

# Read 16 bytes from EEPROM page 0x50 at offset 0
sudo python3 i2c-tools/i2c.py eeprom read --bus 4 --chip 0x50 --offset 0 --
length 16

# Write calibration data to EEPROM
sudo python3 i2c-tools/i2c.py eeprom write --bus 4 --chip 0x50 --offset 0 --
data 0xAA 0xBB 0xCC

# Restore from backup
sudo python3 i2c-tools/i2c.py eeprom restore --bus 4 --in eeprom_backup.bin
```

Which I2C bus to use? (Raspberry Pi 5)

CSI Port	Bus Number
CAM1	`--bus 4`
CAM0	`--bus 6`

Not sure? Run `ls /dev/i2c-*` to list available buses, then use `i2cdetect -y 4` (or `-y 6`) to confirm the camera is present — you should see EEPROM chips at addresses 0x50–0x53.

For detailed usage, sensor register access, and troubleshooting, see `i2c-tools/README.md`.

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5. Testing the Camera

After installation and rebooting, you can verify the camera detection and test its functionality using the standard `rpicam-apps`.

4.1 Verify Detection

List the available cameras to ensure the IMX585 is detected correctly:

```
rpicam-hello --list-cameras
```

4.2 Capture Commands

Live Preview:

```
rpicam-hello -t 0
```

Capture a 4K Image:

```
rpicam-still -o 4k_image.jpg --width 3856 --height 2180
```

Record a 4K Video (30fps):

```
rpicam-vid -t 10000 --width 3856 --height 2180 -o 4k_video.h264
```

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6. Support

For technical support, detailed documentation, and product inquiries, please visit INNO-MAKER.